

BhoomiSetu

GIS-based Land Governance & Interoperability Platform — Smart India Hackathon 2026

SUPPORTING DOCUMENT — BHASHINI API ACCESS REQUEST

1. About the Project

BhoomiSetu is a GIS-based, parcel-centric land governance and interoperability platform built for Smart India Hackathon 2026's "Land Stack" problem statement, which envisions an integrated, map-based Digital Public Infrastructure connecting fragmented land-record datasets — ownership, boundaries, service requests, and inter-departmental workflows — around a single parcel identity. The platform serves three distinct users: citizens searching and claiming land parcels or filing service requests, field officers processing those requests, and administrators managing map layers and monitoring the system, all on top of a live PostGIS-backed spatial database and an interactive map (MapLibre GL JS).

2. Current Localization State

BhoomiSetu already ships a fully bilingual interface (English and Hindi) across every portal — citizen, officer, and admin — using `i18next/react-i18next` with hand-written translation files. This covers navigation, the landing page, parcel search, service-request workflows, and the GIS map's layer labels and popups. However, land governance in India involves citizens far more comfortable in their own regional language than in Hindi or English, and hand-translating every UI string and every dynamically generated notification into all scheduled Indian languages does not scale. This is a known, explicitly tracked gap in the project (further languages are scaffolded in code but blocked on translation content).

3. Intended Use of the Bhashini API

We intend to integrate the Bhashini (Digital India, NLTM) API, server-side, from BhoomiSetu's NestJS backend. We are requesting access to all six Bhashini services:

- **NMT (Neural Machine Translation):** Real-time translation of UI content, service-request status updates, and officer/admin correspondence into the full set of Indian scheduled languages, extending coverage beyond the current English/Hindi pair without hand-maintaining a translation file per language.
- **ASR (Automatic Speech Recognition):** Allowing citizens with limited literacy to file land-related service requests and search for parcels by voice, in their own language.
- **TTS (Text-to-Speech):** Reading back request status, notifications, and key parcel information audibly, in the citizen's chosen language.
- **Transliteration:** Converting owner names, addresses, and parcel-related text between scripts (e.g., Roman to Devanagari/regional scripts) so land-record search stays consistent regardless of the script a citizen types in.
- **OCR (Optical Character Recognition):** Extracting text from scanned land documents and property deeds written in regional-language scripts, complementing BhoomiSetu's existing English-only OCR pipeline (Document Verification feature) with regional-language document support.
- **ALD (Audio Language Detection):** Automatically identifying which language a citizen is speaking so voice input can be routed to the correct ASR/TTS pipeline without asking them to manually select a language first.

This directly serves the problem statement's own goal of accessible, inclusive land governance — reaching rural and semi-literate citizens, not only English/Hindi-literate users on desktop browsers.

4. Technical Integration Summary

Aspect	Detail
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Where the key is used	Web application (NestJS backend service call; never exposed to the browser — same pattern already used for this project's Groq/Gemini AI keys)
Bhashini services requested	All: ASR, NMT, TTS, Transliteration, OCR, ALD
Expected usage tier	Development / prototype — evaluation and demonstration use during SIH 2026, low request volume
Data shared with the API	Non-sensitive UI strings, public parcel/service-request metadata, and citizen-entered request text; no financial, biometric, or identity-document data is sent
Frontend stack	React + TypeScript, Vite, MapLibre GL JS
Backend stack	NestJS (TypeScript), TypeORM, PostgreSQL + PostGIS

5. Declaration

We confirm that the Bhashini API key requested will be used solely for the development, demonstration, and evaluation of the BhoomiSetu prototype as part of Smart India Hackathon 2026, on a non-commercial basis. We agree to use the API within its documented rate limits and terms of service, to keep the API key confidential and server-side, and to inform Bhashini support if the scope of use changes beyond what is described in this document.